Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: B & J Dairy

Physical address of dairy:

9950 Arno RD	Galt	Sacramento	95632
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1958

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X148-X020-X035-XXXX

B. OPERATORS

Pellandini, James

Operator name: Pellandini, James

Telephone no.: (209) 745-2520 Landline, (209) 346-6173 Cellular

10490 Arno RD	Galt	CA	95632
Mailing Address Number and Street	City	State	Zip Code

Pellandini Family Trust, James & Gail

Operator name: Pellandini Family Trust, James & Gail

Telephone no.: (209) 745-2520 Landline, (209) 346-6169 Cellular

10490 Arno RD	Galt	CA	95632
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Pellandini Family Trust, James & Gail

Legal owner name:	Pellandini Family Trust, James & Gail	Telephone no.:	(209) 745-2520	(209) 346-6169
			Landline	Cellular
10490 Arno RD	Galt		CA	95632
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	800	130	0
Number under roof	585	119	500	0	0	100
Maximum number	585	119	500	800	130	100
Average number	585	119	500	800	130	100
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 63 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 29,991.40 *tons per reporting period*

Total nitrogen from manure: 346,754.24 *lbs per reporting period*

After ammonia losses (30% loss applied): 242,727.97 *lbs per reporting period*

Total phosphorus from manure: 56,635.46 *lbs per reporting period*

Total potassium from manure: 109,342.48 *lbs per reporting period*

Total salt from manure: 302,811.30 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,676,000 *gallons*

Total nitrogen generated: 22,553.98 *lbs*

Total phosphorus generated: 3,035.17 *lbs*

Total potassium generated: 39,187.31 *lbs*

Total salt generated: 290,948.76 *lbs*

	11,676,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	11,676,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dairy/Dom.	Ground water
Irr. Well #1	Ground water
Irr. Well #2	Ground water
Irr. Well #3	Ground water
Shop/Beef Cows Dom.	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	Corral solids	910.00 *ton*	Dry-weight	67.9		26,200.00	7,620.00	21,500.00		54.60
11/08/2023	Corral solids	3,555.00 *ton*	Dry-weight	67.9		26,200.00	7,620.00	21,500.00		54.60

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	75,103.09	21,842.96	61,630.40	1,565,125.38
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	75,103.09	21,842.96	61,630.40	1,565,125.38

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1A	7	7	1	process wastewater	X148-X020-X036-XXXX
Field #1B	39	39	2	both	X148-X020-X036-XXXX
Field #1C	12	12	0	none	X148-X020-X035-XXXX
Field #2	72	72	4	both	X148-X020-X036-XXXX
Field #3	48	48	4	both	X148-X020-X036-XXXX
Totals for areas that were used for application	166	166	11		
Totals for areas that were not used for application	12	12	0		
Land application area totals	178	178	11		

B. CROPS AND HARVESTS

Field #1A

Field name: Field #1A

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	240.00 *ton*	Dry-weight		74.4	12,900.00	3,220.00	15,600.00		8.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	43.20	216.00	0.00
Total actual harvest content	34.29	226.45	56.52	273.85	1,548.29

Field #1B

Field name: Field #1B

Field #1B

10/28/2022: Rye/Oat, silage

Crop: Rye/Oat, silage Acres planted: 39 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	507.00 *ton*	Dry-weight		47.2	12,200.00	3,800.00	36,000.00		9.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	180.60	56.00	266.00	0.00
Total actual harvest content	13.00	167.48	52.17	494.21	1,316.52

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,209.00 *ton*	Dry-weight		76.2	12,800.00	3,040.00	15,400.00		7.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	37.80	197.10	0.00
Total actual harvest content	31.00	188.88	44.86	227.24	1,114.08

Field #2

Field name: Field #2

10/28/2022: Rye/Oat, silage

Crop: Rye/Oat, silage Acres planted: 72 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	520.00 *ton*	Dry-weight		43.4	11,900.00	3,900.00	36,000.00		8.38
04/14/2023	416.00 *ton*	Dry-weight		59.6	14,300.00	4,700.00	49,000.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	189.00	61.60	280.00	0.00
Total actual harvest content	13.00	164.05	53.83	523.07	1,212.65

Field #2

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	992.00 *ton*	Dry-weight		74.4	12,900.00	3,220.00	15,600.00		8.82
09/25/2023	1,240.00 *ton*	Dry-weight		77.2	12,700.00	3,150.00	16,500.00		7.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	199.80	0.00
Total actual harvest content	31.00	190.74	47.45	239.63	1,218.25

Field #3

Field name: Field #3

10/28/2022: Rye/Oat, silage

Crop: Rye/Oat, silage Acres planted: 48 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	104.00 *ton*	Dry-weight		59.6	14,300.00	4,700.00	49,000.00		11.30
04/14/2023	520.00 *ton*	Dry-weight		55.8	12,800.00	3,700.00	41,000.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	158.20	44.80	252.00	0.00
Total actual harvest content	13.00	147.62	43.66	478.43	1,327.87

Field #3

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,240.00 *ton*	Dry-weight		75.8	13,400.00	2,940.00	15,200.00		10.60
09/25/2023	248.00 *ton*	Dry-weight		74.6	13,100.00	2,930.00	14,600.00		8.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	199.80	0.00
Total actual harvest content	31.00	201.93	44.45	228.37	1,537.43

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1A - 05/16/2023: Corn, silage

Field name: Field #1A

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	92.13	0.00	0.46	881.23	336,000.00 *gal*
Application event totals		92.13	0.00	0.46	881.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	4.70	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		4.70	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	4.23	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	60.58	4.76	58.43	621.87	126,000.00 *gal*
Irr. Well #1	Ground water	4.23	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		64.81	4.76	58.43	621.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	4.23	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Field #1A - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	60.58	4.76	58.43	621.87	126,000.00 *gal*
Irr. Well #1	Ground water	4.23	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		64.81	4.76	58.43	621.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	4.23	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Field #1B - 10/28/2022: Rye/Oat, silage

Field name: Field #1B

Crop: Rye/Oat, silage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	168.48	32.54	78.48	1,929.60	300.00 *ton*
Application event totals		168.48	32.54	78.48	1,929.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	17.51	6.99	94.04	435.69	1,008,000.00 *gal*
Application event totals		17.51	6.99	94.04	435.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	8.76	3.49	47.02	217.84	504,000.00 *gal*
Application event totals		8.76	3.49	47.02	217.84	

Field #1B - 05/16/2023: Corn, silage

Field name: Field #1B

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	97.95	17.54	74.87	0.00	200.00 *ton*
Application event totals		97.95	17.54	74.87	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	4.22	0.00	0.00	0.00	4,800,000.00 *gal*
Application event totals		4.22	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	3.80	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		3.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	43.49	3.42	41.95	446.47	504,000.00 *gal*
Irr. Well #1	Ground water	3.80	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		47.29	3.42	41.95	446.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	3.80	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		3.80	0.00	0.00	0.00	

Field #1B - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	43.49	3.42	41.95	446.47	504,000.00 *gal*
Irr. Well #1	Ground water	3.80	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		47.29	3.42	41.95	446.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #1	Ground water	3.80	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		3.80	0.00	0.00	0.00	

Field #2 - 10/28/2022: Rye/Oat, silage

Field name: Field #2

Crop: Rye/Oat, silage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	152.10	29.38	70.85	1,742.00	500.00 *ton*
Application event totals		152.10	29.38	70.85	1,742.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	14.23	5.68	76.41	353.99	1,512,000.00 *gal*
Application event totals		14.23	5.68	76.41	353.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	9.49	3.79	50.94	236.00	1,008,000.00 *gal*
Application event totals		9.49	3.79	50.94	236.00	

Field #2 - 05/16/2023: Corn, silage

Field name: Field #2

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	106.11	19.00	81.11	0.00	400.00 *ton*
Application event totals		106.11	19.00	81.11	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #2	Ground water	3.84	0.00	0.00	0.00	8,100,000.00 *gal*
Application event totals		3.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #2	Ground water	3.56	0.00	0.00	0.00	7,500,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	70.68	5.56	68.17	725.51	1,512,000.00 *gal*
Irr. Well #2	Ground water	3.56	0.00	0.00	0.00	7,500,000.00 *gal*
Application event totals		74.23	5.56	68.17	725.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #2	Ground water	3.56	0.00	0.00	0.00	7,500,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Field #2 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	47.12	3.70	45.45	483.68	1,008,000.00 *gal*
Irr. Well #2	Ground water	3.56	0.00	0.00	0.00	7,500,000.00 *gal*
Application event totals		50.67	3.70	45.45	483.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #2	Ground water	3.56	0.00	0.00	0.00	7,500,000.00 *gal*
Application event totals		3.56	0.00	0.00	0.00	

Field #3 - 10/28/2022: Rye/Oat, silage

Field name: Field #3

Crop: Rye/Oat, silage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	136.89	26.44	63.77	1,567.80	300.00 *ton*
Application event totals		136.89	26.44	63.77	1,567.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	14.23	5.68	76.41	353.99	1,008,000.00 *gal*
Application event totals		14.23	5.68	76.41	353.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	14.23	5.68	76.41	353.99	1,008,000.00 *gal*
Application event totals		14.23	5.68	76.41	353.99	

Field #3 - 05/16/2023: Corn, silage

Field name: Field #3

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	99.48	17.81	76.04	0.00	250.00 *ton*
Application event totals		99.48	17.81	76.04	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #3	Ground water	4.01	0.00	0.00	0.00	5,700,000.00 *gal*
Application event totals		4.01	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #3	Ground water	3.55	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	70.68	5.56	68.17	725.51	1,008,000.00 *gal*
Irr. Well #3	Ground water	3.55	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		74.23	5.56	68.17	725.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #3	Ground water	3.55	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Field #3 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	35.34	2.78	34.09	362.76	504,000.00 *gal*
Irr. Well #3	Ground water	3.55	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		38.89	2.78	34.09	362.76	

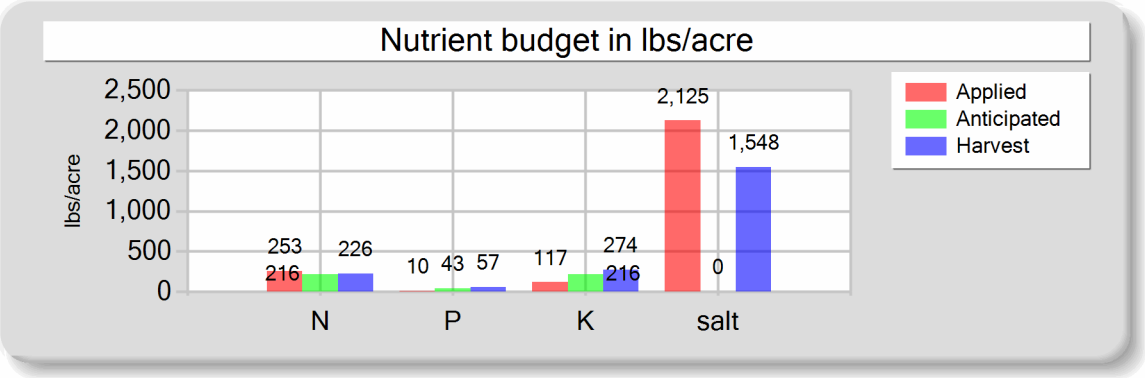
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Well #3	Ground water	3.55	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #1A - 05/16/2023: Corn, silage

Field name: Field #1A Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.29	9.52	117.32	2,124.97
Fresh water	25.87	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	253.16	9.52	117.32	2,124.97
Anticipated crop nutrient removal	216.00	43.20	216.00	0.00
Actual crop nutrient removal	226.45	56.52	273.85	1,548.29
Nutrient balance	26.71	-47.00	-156.52	576.68
Applied to removed ratio	1.12	0.17	0.43	1.37

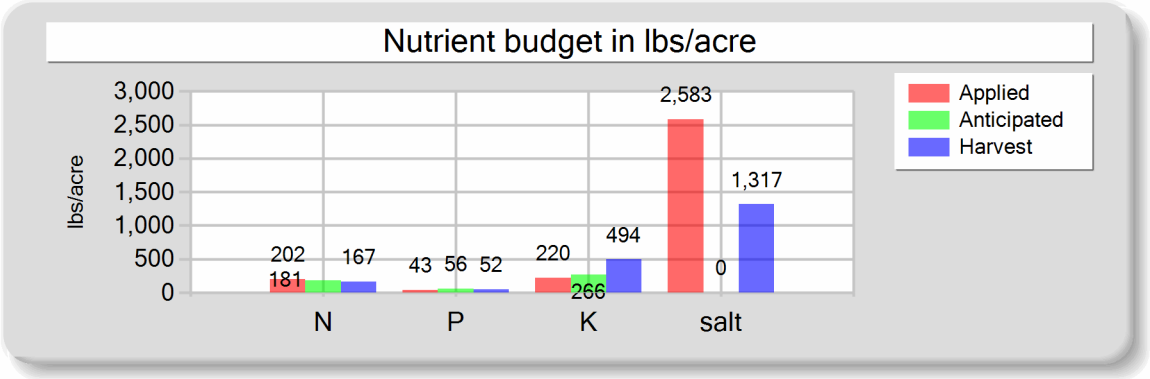
Fresh water applied	
5,280,000.00	*gallons*
194.44	*acre-inches*
27.78	*inches/acre*

Process wastewater applied	
588,000.00	*gallons*
21.65	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #1B - 10/28/2022: Rye/Oat, silage

Field name: Field #1B Crop: Rye/Oat, silage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	168.48	32.54	78.48	1,929.60
Process wastewater	26.27	10.48	141.06	653.53
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.75	43.03	219.54	2,583.13
Anticipated crop nutrient removal	180.60	56.00	266.00	0.00
Actual crop nutrient removal	167.48	52.17	494.21	1,316.52
Nutrient balance	34.27	-9.14	-274.67	1,266.61
Applied to removed ratio	1.20	0.82	0.44	1.96

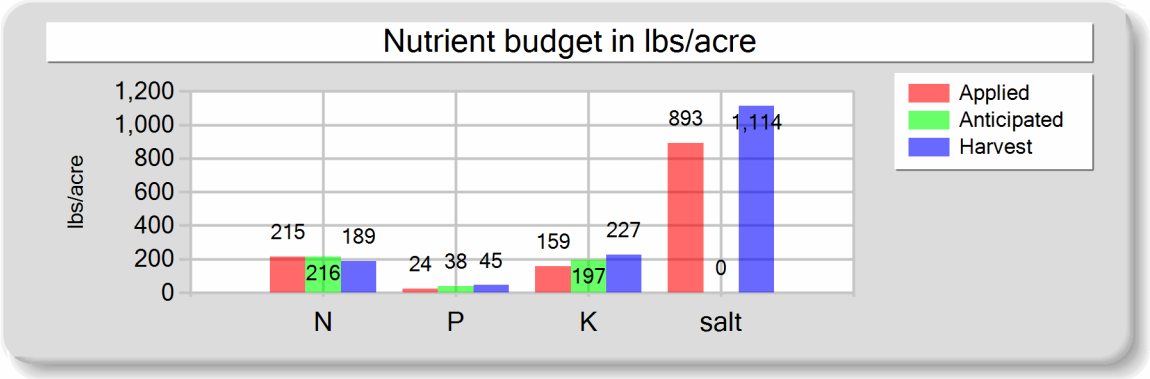
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,512,000.00 *gallons*
55.68 *acre-inches*
1.43 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #1B - 05/16/2023: Corn, silage

Field name: Field #1B Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	97.95	17.54	74.87	0.00
Process wastewater	86.99	6.84	83.90	892.94
Fresh water	23.22	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.15	24.38	158.77	892.94
Anticipated crop nutrient removal	216.00	37.80	197.10	0.00
Actual crop nutrient removal	188.88	44.86	227.24	1,114.08
Nutrient balance	26.28	-20.48	-68.47	-221.14
Applied to removed ratio	1.14	0.54	0.70	0.80

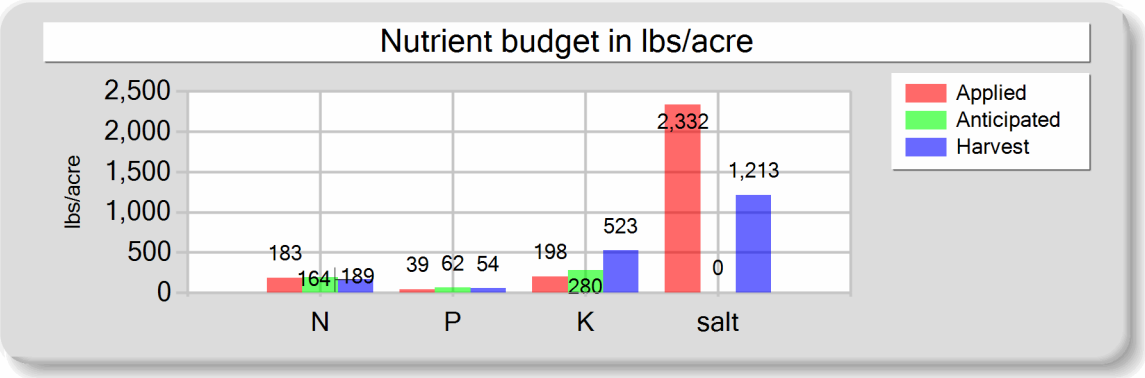
Fresh water applied	
26,400,000.00	*gallons*
972.22	*acre-inches*
24.93	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 10/28/2022: Rye/Oat, silage

Field name: Field #2 Crop: Rye/Oat, silage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152.10	29.38	70.85	1,742.00
Process wastewater	23.72	9.46	127.34	589.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.82	38.84	198.19	2,331.99
Anticipated crop nutrient removal	189.00	61.60	280.00	0.00
Actual crop nutrient removal	164.05	53.83	523.07	1,212.65
Nutrient balance	18.77	-14.98	-324.88	1,119.35
Applied to removed ratio	1.11	0.72	0.38	1.92

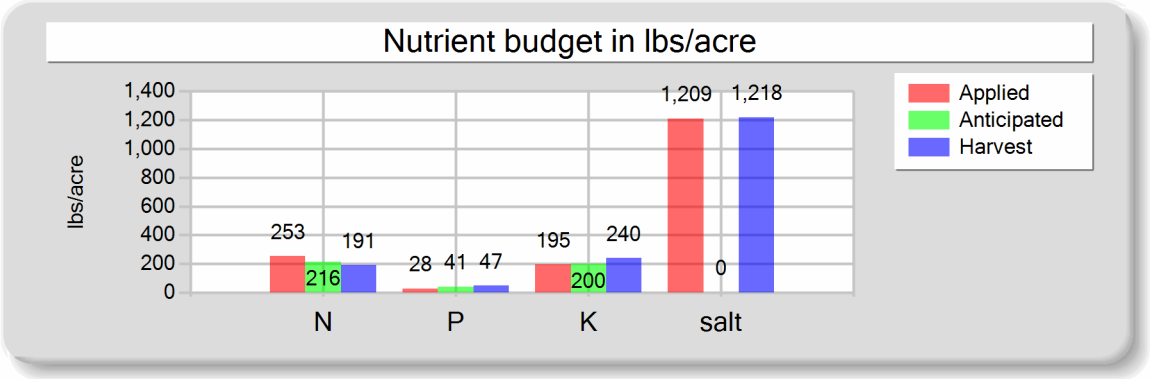
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,520,000.00 *gallons*
92.80 *acre-inches*
1.29 *inches/acre*

Total harvests for the crop
2 *harvests*

Field #2 - 05/16/2023: Corn, silage

Field name: Field #2 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	106.11	19.00	81.11	0.00
Process wastewater	117.80	9.26	113.62	1,209.19
Fresh water	21.62	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.52	28.26	194.73	1,209.19
Anticipated crop nutrient removal	216.00	40.50	199.80	0.00
Actual crop nutrient removal	190.74	47.45	239.63	1,218.25
Nutrient balance	61.79	-19.19	-44.90	-9.06
Applied to removed ratio	1.32	0.60	0.81	0.99

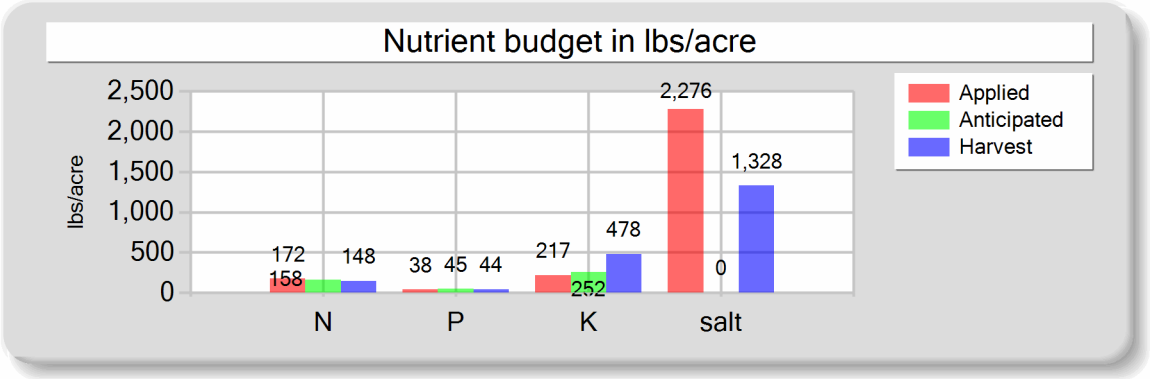
Fresh water applied	
45,600,000.00	*gallons*
1,679.29	*acre-inches*
23.32	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #3 - 10/28/2022: Rye/Oat, silage

Field name: Field #3 Crop: Rye/Oat, silage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	136.89	26.44	63.77	1,567.80
Process wastewater	28.46	11.36	152.81	707.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.35	37.80	216.58	2,275.79
Anticipated crop nutrient removal	158.20	44.80	252.00	0.00
Actual crop nutrient removal	147.62	43.66	478.43	1,327.87
Nutrient balance	24.73	-5.86	-261.85	947.92
Applied to removed ratio	1.17	0.87	0.45	1.71

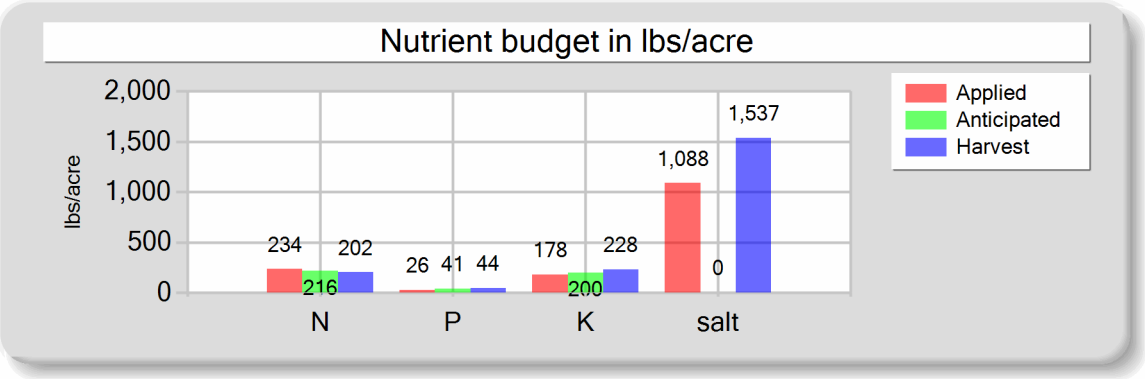
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,016,000.00 *gallons*
74.24 *acre-inches*
1.55 *inches/acre*

Total harvests for the crop
2 *harvests*

Field #3 - 05/16/2023: Corn, silage

Field name: Field #3 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	99.48	17.81	76.04	0.00
Process wastewater	106.02	8.33	102.26	1,088.27
Fresh water	21.76	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.25	26.15	178.30	1,088.27
Anticipated crop nutrient removal	216.00	40.50	199.80	0.00
Actual crop nutrient removal	201.93	44.45	228.37	1,537.43
Nutrient balance	32.33	-18.30	-50.07	-449.15
Applied to removed ratio	1.16	0.59	0.78	0.71

Fresh water applied	
30,900,000.00	*gallons*
1,137.94	*acre-inches*
23.71	*inches/acre*

Process wastewater applied	
1,512,000.00	*gallons*
55.68	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
2	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2313016B

Sample and source description: T2313016B

Sample date: 11/09/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,400.00	4,520.00	10,900.00	36,400.00	6,320.00	2,020.00	3,600.00	2,200.00		26.80
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		0.10

T3139074B

Sample and source description: T3139074B

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	3,420.00	14,600.00							0.00
DL	1.00	1.00	1.00							1.00

T3264045B

Sample and source description: T3264045B

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	7,620.00	21,500.00	29,700.00	13,000.00	3,380.00	9,310.00	1,890.00		54.60
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3059017A

Sample and source description: T3059017A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	81.20	50.40	0.00	0.00	32.40	436.00								2,380.00	2,020
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3139074A

Sample and source description: T3139074A

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	230.00	185.00	0.00	0.00	0.00	1.15								3,380.00	2,200
DL	1.00	1.00	1.00	1.00	0.20	1.00								100.00	1

T3264045A

Sample and source description: T3264045A

Sample date: 09/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	403.00	305.00	0.00	0.31	31.70	389.00	104.00	99.20	70.90	2,000.00	0.00	158.00	139.00	4,850.00	4,140
DL	1.00	1.00	1.00	0.10	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

Dairy/Dom.

Dairy/Dom.

T3307006A

Sample description: T3307006A

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.70	0.00	6.70	35.30	27.80	23.20	268.00	0.00	29.30	21.00	652.00	345
DL	1.00	0.50	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

Irr. Well #1

T3307006C

Sample description: T3307006C

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.11	0.00	4.11								504.00	0
DL	1.00	0.50	1.00								100.00	1

Irr. Well #2

T3307006D

Sample description: T3307006D

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.09	0.00	4.09								503.00	0
DL	1.00	0.50	1.00								100.00	1

Irr. Well #3

Irr. Well #3

T3307006E

Sample description: T3307006E

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.05	0.00	4.05								503.00	0
DL	1.00	0.50	1.00								100.00	1

Shop/Beef Cows Dom.

T3307006B

Sample description: T3307006B

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.82	0.00	0.82								330.00	0
DL	0.50	0.50	0.50								100.00	1

D. SOIL ANALYSES

Field #1A

T3349018A

Sample and source description: T3349018A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			70.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #1B

T3349018A

Sample and source description: T3349018A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			70.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3349018B

Sample and source description: T3349018B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			48.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #2

T3349018B

Sample and source description: T3349018B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			48.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #2

T3349018C

Sample and source description: T3349018C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			49.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #3

T3349018D

Sample and source description: T3349018D

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			58.00		0.00	0.00	
DL			1.00		1.00	1.00	

T349018C

Sample and source description: T349018C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			49.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

Field #1A - 05/16/2023: Corn, silage

Field #1A - 05/16/2023: Corn, silage

T3307006G

Sample and source description: T3307006G

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	3,220.00	15,600.00		8.82
DL	1.00	1.00	1.00		1.00

Field #1B - 10/28/2022: Rye/Oat, silage

W3108008A

Sample and source description: W3108008A

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	3,800.00	36,000.00		9.59
DL	1.00	1.00	1.00		1.00

Field #1B - 05/16/2023: Corn, silage

T3307006F

Sample and source description: T3307006F

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	3,040.00	15,400.00		7.55
DL	1.00	1.00	1.00		1.00

Field #2 - 10/28/2022: Rye/Oat, silage

W3108008B

Sample and source description: W3108008B

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	3,900.00	36,000.00		8.38
DL	1.00	1.00	1.00		1.00

W3108008C

Sample and source description: W3108008C

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	4,700.00	49,000.00		11.30
DL	1.00	1.00	1.00		1.00

Field #2 - 05/16/2023: Corn, silage

T3307006G

Sample and source description: T3307006G

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	3,220.00	15,600.00		8.82
DL	1.00	1.00	1.00		1.00

Field #2 - 05/16/2023: Corn, silage

T3307006H

Sample and source description: T3307006H

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	3,150.00	16,500.00		7.59
DL	1.00	1.00	1.00		1.00

Field #3 - 10/28/2022: Rye/Oat, silage

W3108008C

Sample and source description: W3108008C

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	4,700.00	49,000.00		11.30
DL	1.00	1.00	1.00		1.00

W3108008D

Sample and source description: W3108008D

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	3,700.00	41,000.00		11.80
DL	1.00	1.00	1.00		1.00

Field #3 - 05/16/2023: Corn, silage

Field #3 - 05/16/2023: Corn, silage

T3307006I

Sample and source description: T3307006I

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,940.00	15,200.00		10.60
DL	1.00	1.00	1.00		1.00

T3307006J

Sample and source description: T3307006J

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,930.00	14,600.00		8.08
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

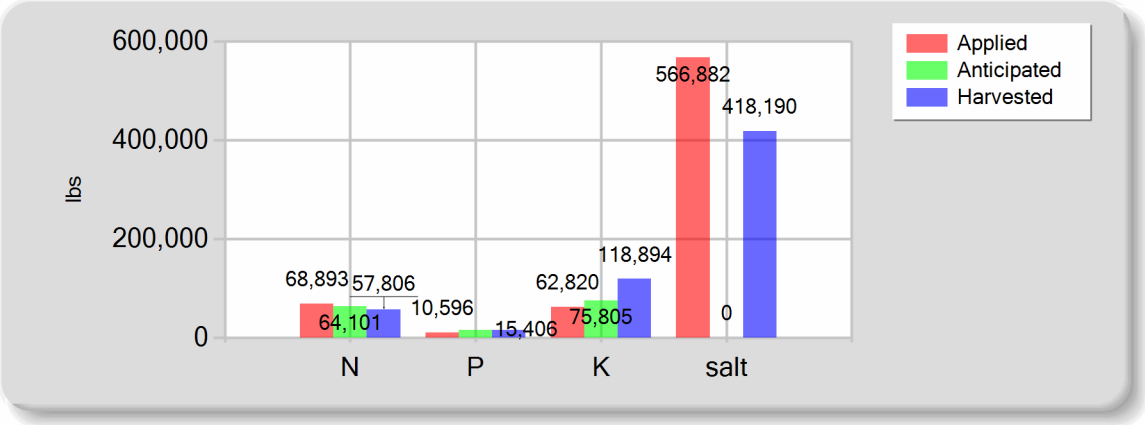
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

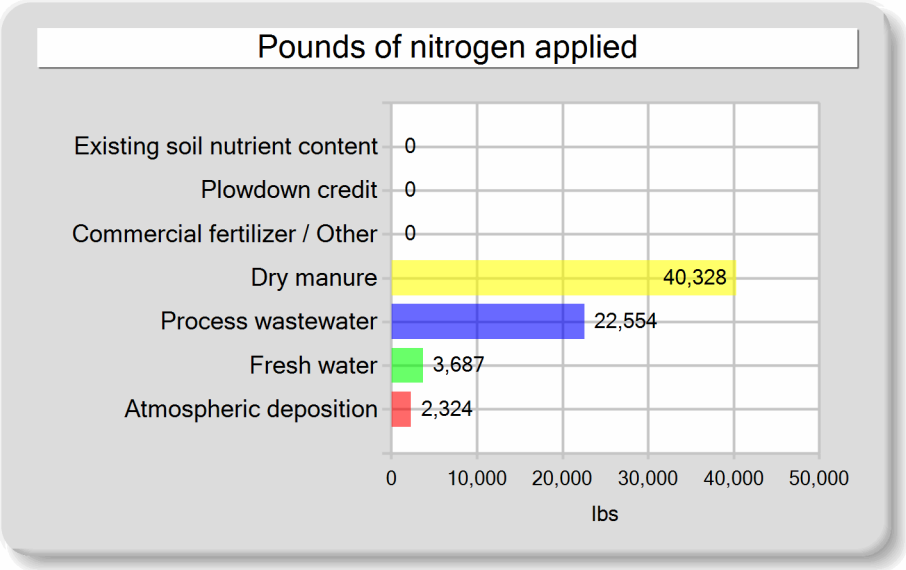
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

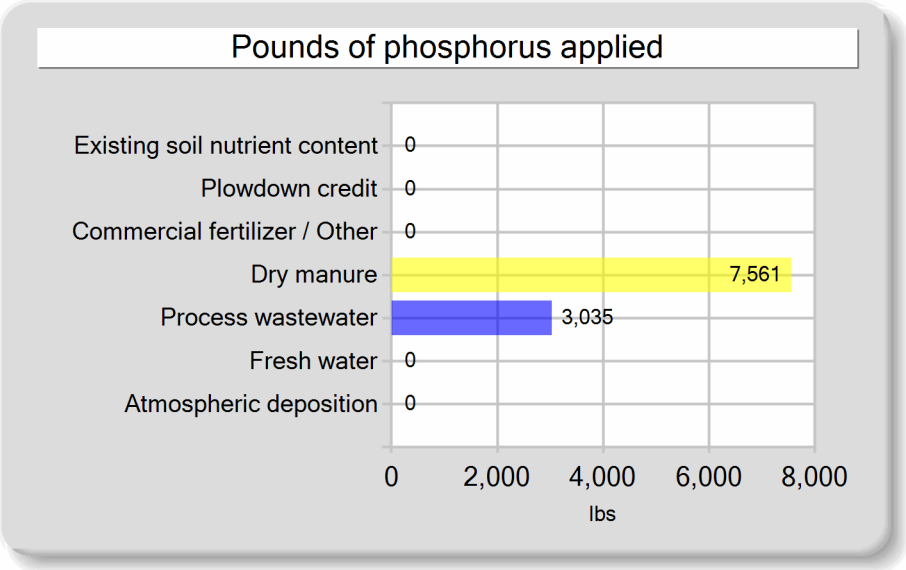
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	40,327.64	7,560.79	23,632.64	275,932.80
Process wastewater	22,553.98	3,035.17	39,187.31	290,948.76
Fresh water	3,687.27	0.00	0.00	0.00
Atmospheric deposition	2,324.00	0.00	0.00	0.00
Total nutrients applied	68,892.89	10,595.97	62,819.95	566,881.56
Anticipated crop nutrient removal	64,101.00	15,406.20	75,804.90	0.00
Actual crop nutrient removal	57,805.72	15,701.09	118,894.11	418,190.00
Nutrient balance	11,087.17	-5,105.12	-56,074.17	148,691.56
Applied to removed ratio	1.19	0.67	0.53	1.36

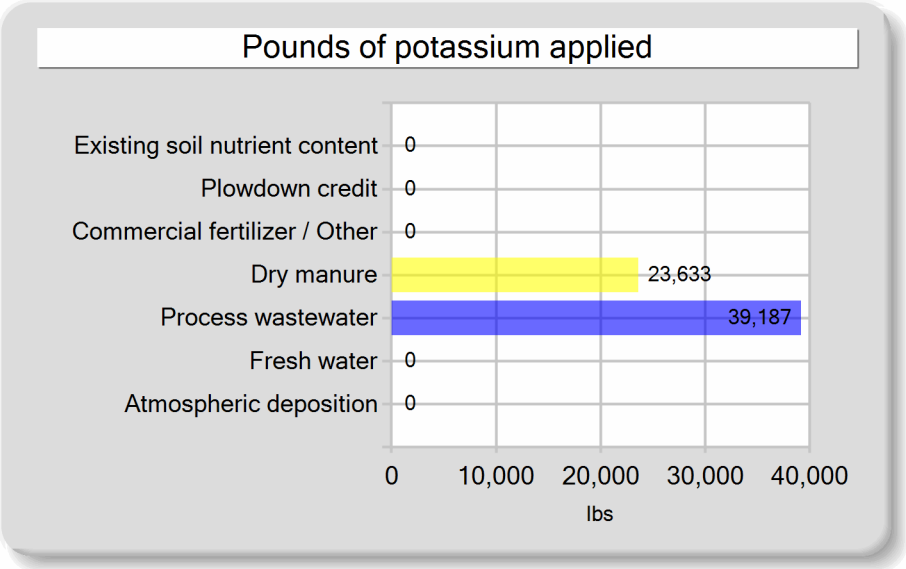
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

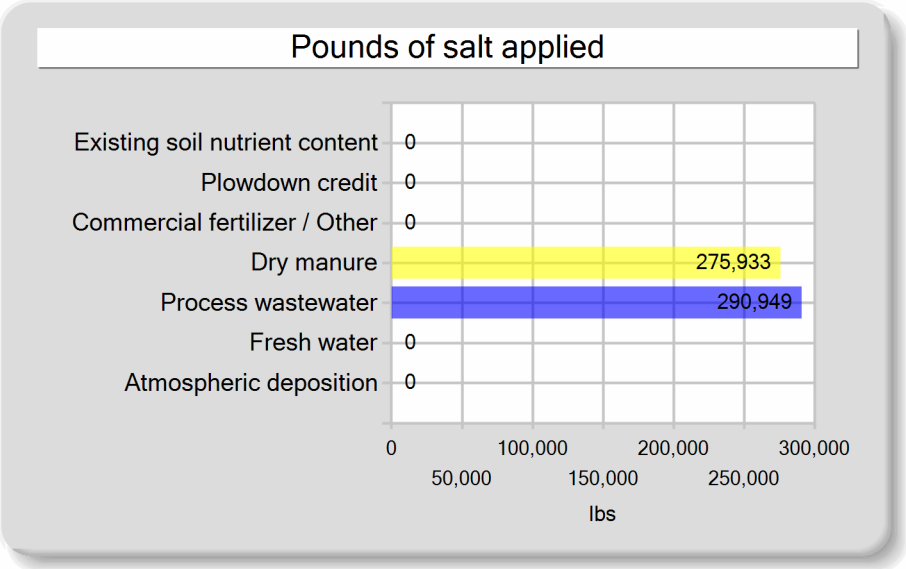


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
James & Gail Pellandini Family Trust	James Pellandini
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-4-2024	6-4-2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Jim Pellandini_____

Name of Dairy Facility: __B & J Dairy_____

Facility Address: _9950 Arno Rd._____Galt_____95632__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Jim Pellandini_____209-346-6173
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Jerry Warren_____

Address of Hauling Company /Person: _27613 Sowles Rd._____Galt_____95632_
Number and Street City Zip Code

Contact Person: ___Jerry Warren_____209-327-3362_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_Ron Silva_____11211 Valensin Rd._____Galt_____95632____510-599-9033_
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

___10080 Arno Rd._____Galt_____95632_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____October-November 3, 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 2.62%
P 0.762% K 2.15%

Manure: ___910_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___67.9%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: hauler's estimate

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: James Pellard Date: 1/23/24

Hauler's Signature: Jim Wan Date: 1/23/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Jim Pellandini_____

Name of Dairy Facility: __B & J Dairy_____

Facility Address: _9950 Arno Rd._____Galt_____95632__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Jim Pellandini_____209-346-6173
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Jerry Warren_____

Address of Hauling Company /Person: _27613 Sowles Rd._____Galt_____95632_
Number and Street City Zip Code

Contact Person: ___Jerry Warren_____209-327-3362_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_Mancebo Ranch_____700 Village Dr._____Galt_____95632____209-479-6360_
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____X134-0280-X086-0000, X134-0280-X044-0000, X134-0280-X043-0000_
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____October-November 8, 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 2.62%
P 0.762% K 2.15%

Manure: ___3,555_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___67.9%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: hauler's estimate

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: James Pellard Date: 1/23/24

Hauler's Signature: [illegible] Date: 1/23/24

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: B & J DAIRY
Mailing address: 10490 ARNO RD
(not facility address) Galt, CA 95632
Phone#: (209) 346-6169

Authorized Copy Released To:
JCRK, INC.
(209) 327-0992
Sampled By: JIM CAPTEIN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	11/2/23	11:30a	Dairy/Dom	Dom. Well	16.8°c	20 min					H14
2	✓	12:00p	Shop/Beef Cows Dom	✓	15.8°	✓	X				
3	✓	12:30p	Irr. Well #1	Ag Well	17.0°	✓	X				
4	✓	1:00p	Irr. Well #2	✓	16.6°	✓	X				
5	✓	1:30p	Irr. Well #3	✓	16.8°	✓	X				
6	10/31/23	10:00a	Field 1B	Corn Silage						X	
7	✓	10:15a	Fields 1A & 2	✓						X	
8	✓	10:30a	Field 2	✓						X	
9	✓	10:45a	Field 3 North	✓						X	
10	✓	11:00a	Field 3 South	✓						X	

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK, INC.		11/3/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-3-23 1:00			T3307006

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
B & J Dairy #1
10490 Arno Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/3/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/3/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy/Dom.	T3307006A	Electrical Conductivity (EC)	EPA 120.1 and	0.652	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	268	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	29.3	mg/L
		Sodium (Na)	EPA 200.7	23.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	27.8	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.70	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	345	mg/L
		Calcium (Ca)	EPA 200.7	35.3	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	21.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	417	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)			2.48
Calcium (Ca)	102			29.2 (W3306004A)	29.1 (DUP)	0.549
Chloride (Cl)	101	106 (W3306004A)	107 (W3306004A)			1.50
Magnesium (Mg)	97.6			21.3 (W3306004A)	21.2 (DUP)	0.235
Nitrate Nitrogen (NO3-N)	99.2	102 (W3306004A)	101 (W3306004A)			1.17
Sodium (Na)	101			31.0 (W3306004A)	30.7 (DUP)	0.843
Sulfate (SO4)	101	105 (W3306004A)	105 (W3306004A)			0.024
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
B & J Dairy #1
10490 Arno Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/3/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/3/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Shop/Beef Cows Dom.	T3307006B	Electrical Conductivity (EC)	EPA 120.1 and	0.330	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.815	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	211	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)	2.48
Electrical Conductivity (EC)	98.0			0.943
Nitrate Nitrogen (NO3-N)	99.3	103 (T3306005A)	101 (T3306005A)	2.39

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Grower

Consultant:
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P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/3/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/3/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 17.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irr. Well #1	T3307006C	Electrical Conductivity (EC)	EPA 120.1 and	0.504	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.11	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	322	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)	2.48
Electrical Conductivity (EC)	98.0			0.943
Nitrate Nitrogen (NO3-N)	99.3	103 (T3306005A)	101 (T3306005A)	2.39

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Consultant:
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P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/3/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/3/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irr. Well #2	T3307006D	Electrical Conductivity (EC)	EPA 120.1 and	0.503	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.09	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	322	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)	2.48
Electrical Conductivity (EC)	98.0			0.943
Nitrate Nitrogen (NO3-N)	99.3	103 (T3306005A)	101 (T3306005A)	2.39

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PURCHASE ORDER:
RECEIVED DATE: 11/3/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/3/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irr. Well #3	T3307006E	Electrical Conductivity (EC)	EPA 120.1 and	0.503	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.05	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	322	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.6	95.8 (Method BI)	94.0 (Method BI)	1.89
Electrical Conductivity (EC)	98.0			0.943
Nitrate Nitrogen (NO3-N)	99.3	103 (T3306005A)	101 (T3306005A)	2.39

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